

Practice problems #20

1. Determine the values of the parameter s for which the given system has a unique solution, and then describe the solution using Cramer's Rule.

$$\begin{cases} sx_1 + 2sx_2 &= -1 \\ 3x_1 + 6sx_2 &= 4 \end{cases}$$

2. Compute the adjugate of the given matrix, and the inverse of the matrix in terms of adjugate.

$$\begin{bmatrix} 3 & 6 & 7 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

3. True or False. Justify your answer.

Suppose that all the entries in A are integers and $\det A = 1$. Then, all the entries in A^{-1} are integers.

Answers or Hints on page 2.

$$\#1: s \neq 0, 1; \quad x_1 = \frac{-7}{3(s-1)}, \quad x_2 = \frac{4s+3}{6s(s-1)}.$$

$$\#2: \operatorname{adj} A = \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}, \quad A^{-1} = (-1) \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

#3: True. (Hint: True for 2×2 case? How about in..... proof?)